

TABLEAU DASHBOARD REPORT

E-Commerce Sales Dataset Day 5

7-Day Data Analyst Portfolio Roadmap Full Step-by-Step Documentation

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1. Introduction

This report documents the process of building an interactive dashboard using Tableau Public, based on the e-commerce sales dataset cleaned on Day 1 (cleaned_sales_data.csv, 3,933 rows). This stage is the visual culmination of all the analysis carried out in Day 2 (Excel), Day 3 (SQL), and Day 4 (Python EDA) turning those findings into a single dashboard that anyone can explore interactively, without needing to open Excel, SQL, or Python.

Like the previous reports, this one is written as a step-by-step tutorial each section explains HOW to do something in Tableau Public (which field to drag, which shelf to use, which menu to click), not just the end result.

Working file: a Tableau dashboard titled "E-Commerce Sales Performance Dashboard", published via Tableau Public.

Link : <https://public.tableau.com/app/profile/alya.shandy/viz/E-CommerceSalesPerformanceDashboard/17876681702130/E-CommerceSalesPerformanceDashboard>

2. Executive Summary

Table 1 summarizes the main components of the dashboard that was built.

Metric	Value
Number of KPI Cards	4 (Total Revenue, Total Orders, Total Customers, AOV)
Number of Charts	5 (Monthly Trend, Revenue by Category, Revenue by Region, Top 10 Products, Orders by Category)
Number of Interactive Filters	3 (Category, Region, Date/Order Date)
Total Revenue (Gross, all orders)	\$306,842
Total Orders (Gross)	3,933
Total Customers	889
Average Order Value (AOV)	\$78.02

Table 1. Summary of Day 5 dashboard components

Note: the figures on this dashboard count all orders (Completed, Cancelled, Returned, Pending) since no status filter is applied by default unlike the Day 3 SQL analysis, which filtered for Completed orders only. This

difference is intentional so the dashboard can be freely explored by the user, including viewing non-Completed data when needed.

3. Analysis Stages (Tutorial)

3.1 Connecting Data to Tableau

1. Open Tableau Public, in the Connect panel (left) choose "Text File" under "To a File" (since the data source is a CSV).
2. Browse and select the cleaned_sales_data.csv file.
3. Once loaded, verify the row and column count at the bottom of the screen it should show "12 fields 3933 rows", matching the Day 1 cleaning result.

3.2 Building 4 KPI Cards

KPI cards are built by changing the Marks type to "Text", then dragging the desired measure into the Text box producing a large number display with no chart axes.

1. Total Revenue: drag the Revenue field into the Text box in Marks. Format the number as Currency (Custom) with a "\$" prefix.
2. Total Orders: drag the Order Id field into the Text box, making sure the aggregation is Count Distinct (CNTD) so it counts unique orders.
3. Total Customers: drag the Customer Id field into the Text box, with Count Distinct aggregation counting unique customers (889), not the row count.
4. Average Order Value (AOV): built via a new Calculated Field with the formula $\text{SUM}([\text{Revenue}]) / \text{COUNTD}([\text{Order Id}])$, then dragged into the Text box and formatted as Currency.

Total Revenue	Total Orders	Total Customers	Average Order Value
\$306,842.20	3,933	889	78.02

Figure 2. The four final KPI cards Total Revenue, Total Orders, Total Customers, Average Order Value

3.3 Chart 1 Monthly Revenue Trend

1. Drag the Order Date field into the Columns shelf, then expand it to the Month level (choose the "Month" option that displays the format "May 2015", not just "May", so the monthly trend stays chronologically ordered across years).
2. Drag the Revenue field into the Rows shelf Tableau automatically forms a line chart.
3. Clean up the display: change the size dropdown from "Standard" to "Fit Width" so all 24 month labels are visible, and set a Fixed Axis so the Y axis doesn't start too far from the data.

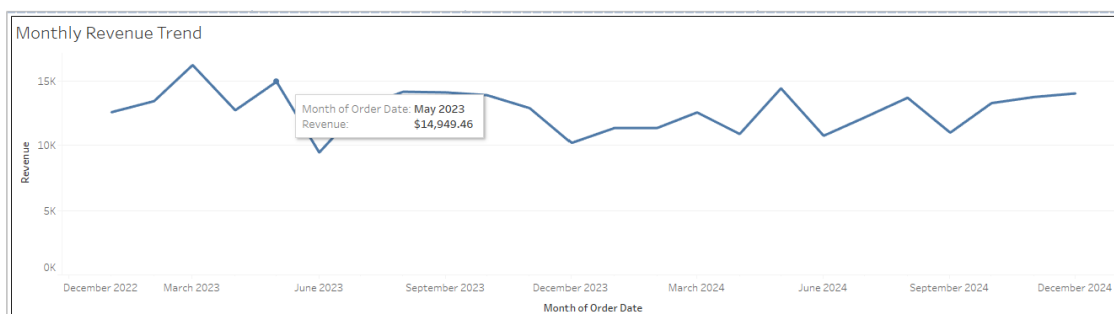


Figure 3. Monthly Revenue Trend a sharp up-and-down pattern across 2023-2024, with the tooltip showing detail per point (example: May 2023 = \$14,949.46)

3.4 Chart 2 Revenue by Category

1. Drag the Category field into the Rows shelf, drag the Revenue field into the Columns shelf forming a horizontal bar chart.
2. Sort in Descending order by Revenue so the highest-revenue category appears at the top.

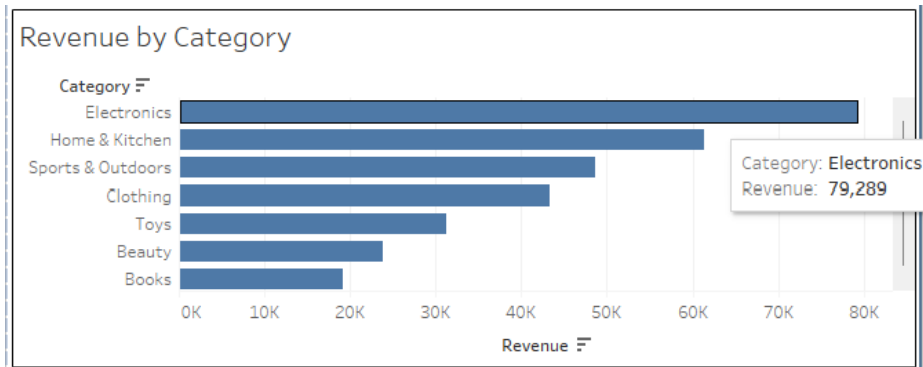


Figure 4. Revenue by Category Electronics leads with \$79,289, Books sits at the bottom

3.5 Chart 3 Revenue by Region

1. Drag the Region field into the Rows shelf, drag the Revenue field into the Columns shelf, then sort descending.

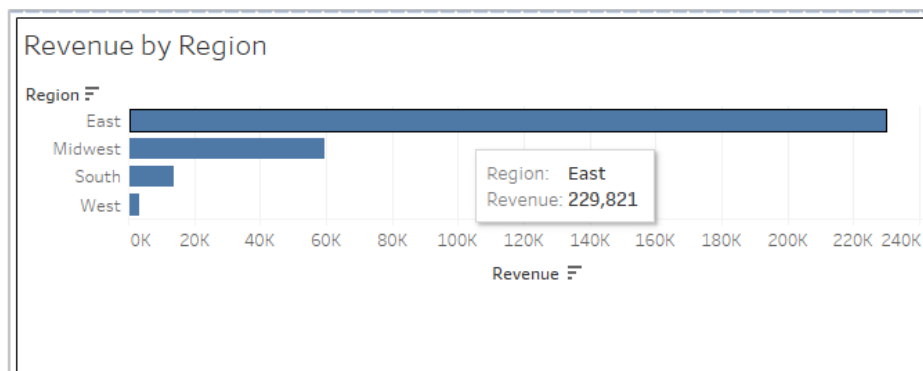


Figure 5. Revenue by Region the East region dominates with \$229,821 in revenue

3.6 Chart 4 Top 10 Products

1. Drag the Product Name field into Rows, Revenue into Columns, then sort descending.
2. Limit to only the top 10 products: right-click the Product Name pill in Rows → Filter → the "Top" tab → select "By field", enter 10, field Revenue, aggregation Sum.

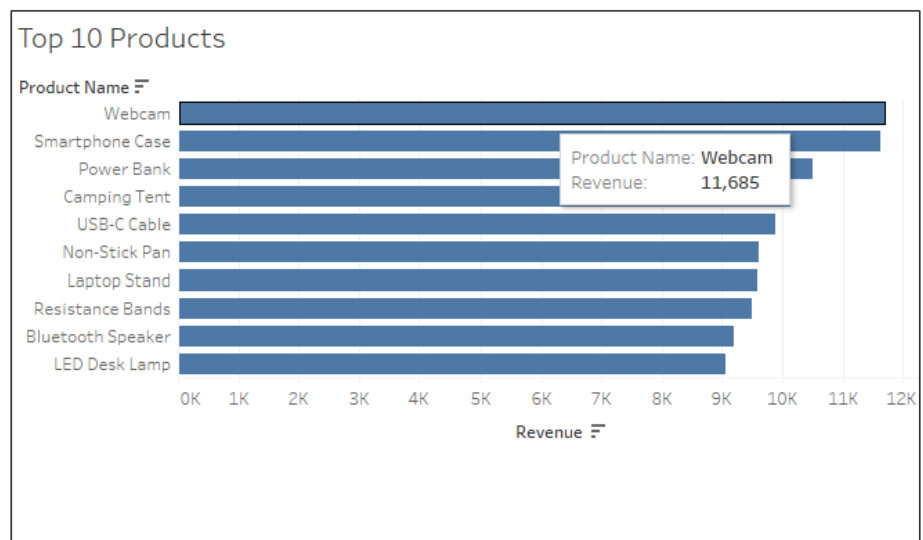


Figure 6. Top 10 Products Webcam leads with 11,685 in revenue (counting all orders, not just Completed)

3.7 Chart 5 Orders by Category

1. Drag the Category field into Rows. Drag the Order Id field into Columns since Order Id is recognized as a Dimension (not a Measure), this field first needs to be changed via right-click → Measure → Count (Distinct), so it's calculated as CNTD(Order Id).
2. Sort descending by Distinct Count of Order Id.

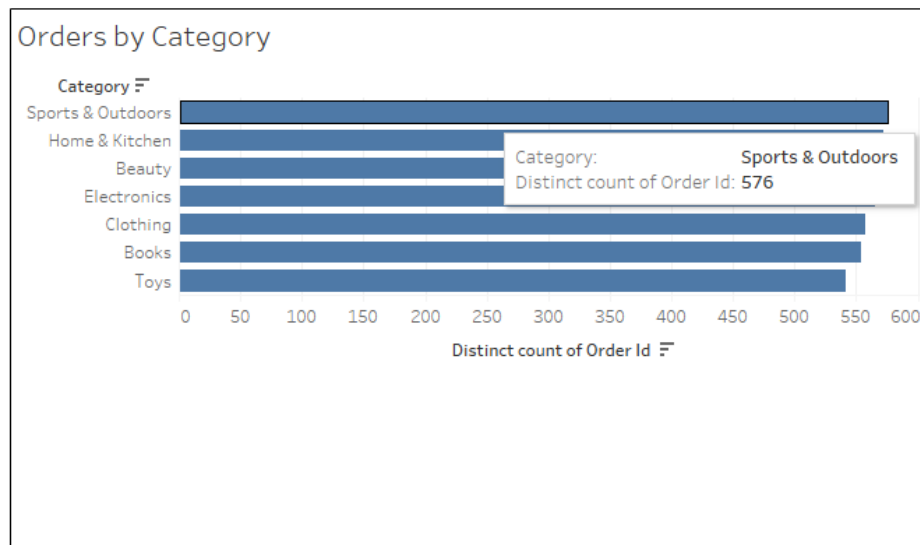


Figure 7. Orders by Category Sports & Outdoors has the highest order volume (576 orders)

Important technical note: ID-type fields (such as Order Id, Customer Id) are often recognized by Tableau as Dimensions/text rather than Measures/numbers so their aggregation needs to be manually changed to Count Distinct every time they're used to count unique records.

3.8 Assembling the Dashboard

1. Click "New Dashboard" to create a new dashboard canvas.
2. Set the dashboard size to Fixed Size "Desktop Browser (1000 x 800)" so it's proportional for screenshots and publishing.
3. Drag all 4 KPI cards from the Sheets panel onto the canvas, arranging them side by side in the top row.
4. Drag the 5 charts one by one into the space below the KPI cards. Tableau automatically arranges the layout (Tiled layout) based on the drop position watch for the blue guide line before releasing the mouse, to make sure a chart doesn't end up nested inside another chart.

3.9 Adding Interactive Filters

Filters are added at the Dashboard level (not per worksheet) so that a single filter can control all charts at once.

1. Category filter: hover over a chart that has the Category field (e.g. Revenue by Category) → click the dropdown icon (▼) in the top-right corner of the chart → Filters → select Category. A checklist-style filter card automatically appears on the dashboard.
2. Region filter: repeat the same steps on a chart that has the Region field (e.g. Revenue by Region) → Filters → Region.
3. Date filter: repeat on the Monthly Revenue Trend chart → Filters → Order Date (Month). Tableau automatically provides a "Range of Dates" filter with a slider from January 2023 to December 2024.

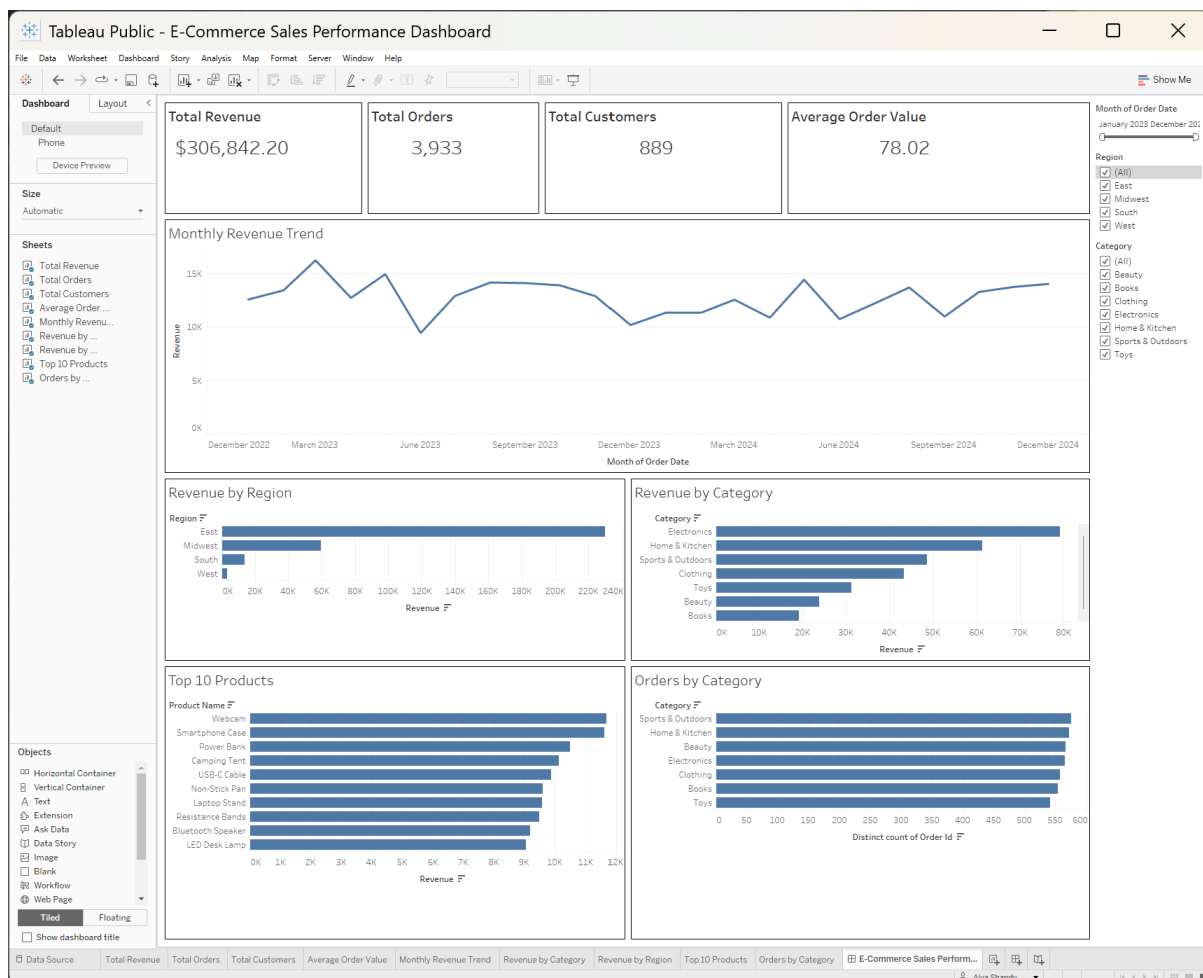


Figure 8. The final dashboard 4 KPI cards, 5 charts, and 3 interactive filters (Date, Region, Category) fully assembled

These three filters are independent of one another and can be combined for example, narrowing to Region "East" and a specific Category at the same time to see that segment's specific performance.

3.10 Reviewing the Layout Testing Filter Interactivity

Before being considered complete, the dashboard was tested by activating a specific filter combination to verify that all KPI cards and other charts update automatically and consistently confirming the dashboard is genuinely interactive, not just a collection of static images. As an example, the Region filter was set to "East" and the Category filter was set to "Books" + "Electronics":

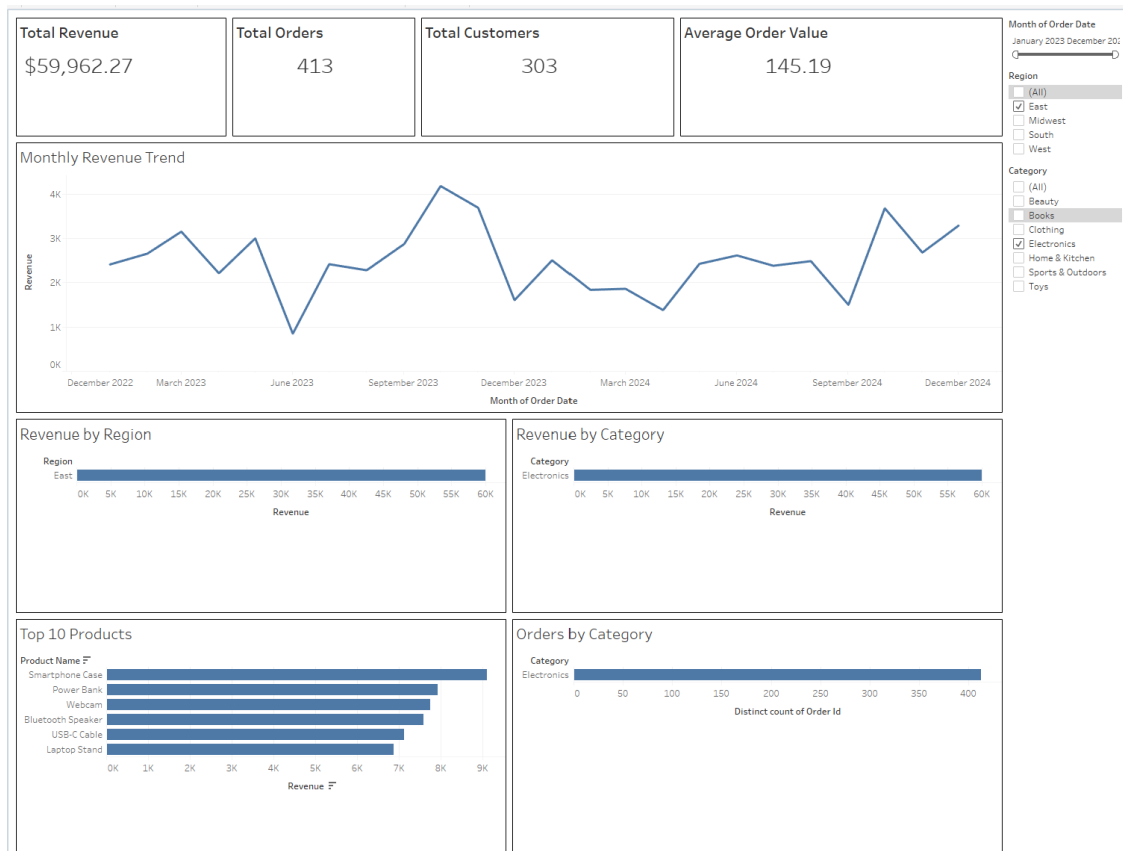


Figure 9. Result of Region = East, Category = Books & Electronics filter all KPIs and charts adjust automatically (Total Revenue drops from \$306,842 to \$59,962, Total Orders from 3,933 to 413)

This test confirms the dashboard responds to filter changes in real time and consistently across every component meeting the standard of a company-ready dashboard fit for decision-making.

4. Business Insights

The insights below summarize what sets a dashboard approach apart from the static analysis done in Day 1-4.

1. The Dashboard Re-Confirms Findings Across Every Stage

Every chart on this dashboard the East region's dominance, Electronics as the strongest category, the low AOV produces the exact same ranking and pattern as the findings in Excel (Day 2), SQL (Day 3), and Python (Day 4), even though it uses gross data (all order statuses) rather than Completed only. This consistency across four different tools reinforces the reliability of every insight found so far.

2. Interactivity Changes How Data Is Consumed

Unlike a static report (PDF, slide, screenshot), the Category/Region/Date filters let anyone explore the data according to their own question for example, a regional manager can immediately filter down to their own region without needing to request a custom report.

3. ID-Type Fields Need Special Handling in BI Tools

The repeated experience of changing Order Id from Dimension to Measure (Count Distinct) across several charts is an important reminder: BI tools like Tableau guess data types automatically, and that guess doesn't always match the intended analysis manual verification is still required.

4. Layout Is Part of the Analysis, Not Just Aesthetics

The process of arranging 4 KPI cards side by side and 5 charts in a proportional grid shows that a good dashboard requires thinking through visual hierarchy (KPIs on top as a quick summary, detail below) not just stacking every chart without structure.

5. Day 5 Checklist Summary

The table below shows the completion status of each Day 5 stage, taken from the portfolio progress tracker.

No	Day	Stage	Task	Tools	Status
46	Day 5	Tableau Dashboard	KPI cards: Revenue, Orders, Customers, AOV	Tableau	DONE
47	Day 5	Tableau Dashboard	Chart: Monthly Revenue Trend	Tableau	DONE
48	Day 5	Tableau Dashboard	Chart: Revenue by Category	Tableau	DONE
49	Day 5	Tableau Dashboard	Chart: Revenue by Region	Tableau	DONE
50	Day 5	Tableau Dashboard	Chart: Top 10 Products	Tableau	DONE
51	Day 5	Tableau Dashboard	Chart: Orders by Category	Tableau	DONE
52	Day 5	Tableau Dashboard	Filter: Date	Tableau	DONE
53	Day 5	Tableau Dashboard	Filter: Region	Tableau	DONE
54	Day 5	Tableau Dashboard	Filter: Category	Tableau	DONE
55	Day 5	Tableau Dashboard	Review layout like a company dashboard	Tableau	DONE
56	Day 5	Tableau Dashboard	Export dashboard screenshot	Tableau	DONE

Table 2. Status of all Day 5 Tableau Dashboard tasks

6. Conclusion

Day 5 produced one interactive Tableau dashboard titled "E-Commerce Sales Performance Dashboard" consisting of 4 KPI cards, 5 charts, and 3 interconnected filters (Category, Region, Date). This dashboard is the visual culmination of the entire Day 1-4 analysis series: data cleaned in Excel, analyzed via pivot tables and SQL, re-validated via Python EDA, now presented in a single view that anyone can explore directly without opening any technical tools.

The key findings most clearly visualized: dependence on the East region, Electronics category dominance, and the gap between Average Order Value and average revenue per customer these three points will be the core discussion in Day 6 (Business Case Study), which will formulate concrete business recommendations based on all the insights gathered throughout Day 1-5.

The dashboard has been published to Tableau Public, and its public link is included in the GitHub repository README, so it can be accessed and explored directly by anyone reviewing this portfolio.